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Atlantic Technological University

ATLANTIC TECHNOLOGICAL UNIVERSITY

WINTER EXAMINATIONS 2025/2026

MODULE: MATHEN705 - Mathematics 5

PROGRAMME(S):

LY_MBIOM_B	Bachelor of Engineering (Honours) in Biomedical Design
LY_MELEC_D	Bachelor of Engineering in Electronic Engineering
LY_CBURE_D	BEng in Building Engineering with Renewable Energy
LY_CCIVL_D	Bachelor of Engineering in Civil Engineering
LY_MEVEN_B	BEng (Honours) in Electric Vehicle Engineering
LY_MMECH_D	Bachelor of Engineering in Mechanical Engineering
LY_CFSTY_B	Bachelor of Engineering (Honours) in Fire Safety Engineering
LY_CBUFR_D	Bachelor of Engineering in Fire Safety Engineering
LY_MECHN_B	Bachelor of Engineering (Honours) in Mechanical Engineering
LY_MELBS_B	Bachelor of Engineering (Honours) in Electronic Engineering

YEAR OF STUDY: 3

EXAMINER(S):

Paddy Hannigan	(Internal)
Dermot Clarke	(External)
Gerald Gallagher	(External)
Dr. Philip Vance	(External)
Dr. Brian Solan	(External)
Mr. Keith Vaugh	(External)
Mr. Brian Gallagher	(External)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer 4 questions. All questions carry equal marks.**

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Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper:
1. Calculator
2. Graph Paper
3. Log Tables

Question 1.

(a) Prove the following hyperbolic identities

$$(i)\ \sinh(2x) = 2\sinh x \cosh x$$
$$(ii)\ \cosh^2 x + \sinh^2 x = \cosh 2x.$$

[7 marks]

(b) Use Osborne's rule to determine the equivalent hyperbolic form of the following trigonometric identities (briefly justify your answers):

$$\begin{aligned}
\text{(i)}\quad \cos(3\theta) &= -\cos^3\theta - 3\sin^2\theta\cos\theta, \\
\text{(ii)}\quad \cos^4\theta &= \frac{1}{8}\left[3 + 4\cos(2\theta) + \cos(4\theta)\right], \\
\text{(iii)}\quad \cos A - \cos B &= -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right), \\
\text{(iv)}\quad \sin(2x) + \sin(2y) + sin(2z) &= 4\sin x \sin y \sin z.
\end{aligned}$$

[10 marks]

(c) Determine if the following functions are odd, even, or neither even nor odd:

$$(i) f(x) = (x^2+1)\sin x, \quad (ii) f(x) = \frac{x}{x^2+x},$$
$$(iii) f(x) = x^2 cos(3x) \quad (iv) f(x) = \sin(\cos(x^3)).$$

In all cases give full justification for your answers.

[8 marks]

Question 2.

(a) Show that the solution to the differential equation

$$\frac{dy}{dx} - y\sin x = y \quad \text{where } y(0) = 1;$$

using separation of variables is

$$y(x) = \mathrm{e}^{-\cos x + x + 1}.$$

[13 marks]

(b) Use the integrating factor method, to show that the solution to the differential equation

$$\frac{dy}{dx} - \frac{1}{x}y = x\sin x, ,$$

is

$$y(x) = -x\cos x + Cx.$$

[12 marks]

Question 3.

(a) Use the Newton-Raphson method to approximate the only positive root of the function

$$f(x) = \cosh x + x - 2$$

accurate to 3 decimal places. Comment on the rate of convergence of this method.

[8 marks]

(b) Outline how part (a) could be solved (the solution is not required) using the Secant Method and any advantages this method has over the Newton-Raphson method for some root finding problems.

[6 marks]

(c) Show that the Taylor series expansion of $f(x) = \sqrt{x}$ in powers of $x - 1$ is

$$\sqrt{x} = 1 + \frac{1}{2}(x-1) - \frac{(x-1)^2}{8} + \frac{(x-1)^3}{16} + \dots$$

[7 marks]

(d) Use the four terms from part (c) to approximate $\sqrt{1.4}$.

[4 marks]

Question 4.

(a) Using the Improved Euler method approximate the solution to the initial value problem

$$\frac{dy}{dx} = y\mathrm{e}^x, \quad \text{where } y(1) = 2,$$

on the interval $1 < x \leq 1.5$, where the step size is $h = 0.25$.

[14 marks]

(b) Approximate the solution to the initial value problem

$$\frac{dy}{dx} = \mathrm{e}^x - y, \quad \text{where } y(1) = 0,$$

at $x = 1.25$ using the Runge-Kutta method of order 4.

[11 marks]

Question 5.

(a) Using the method of Frobenius, show that the differential equation

$$\frac{d^2y}{dx^2} + \frac{1}{2x}\frac{dy}{dx} + \frac{1}{4x}y = 0,$$

has indicial equation

$$c(c - \frac{1}{2})a_0 = 0.$$

[9 marks]

(b) Show, using the method of Frobenius, that the solution to the differential equation in part (a) takes the form

$$\begin{aligned} y(x) &= A\{1 - \frac{x}{2!} + \frac{x^2}{4!} - \frac{x^3}{6!} + \ldots\} \\ & \quad B\sqrt{x}\{1 - \frac{x}{3!} + \frac{x^2}{5!} - \frac{x^3}{7!} + \ldots\}. \end{aligned}$$

[16 marks]

Question 6.

(a) Show that the eigenvalues of the matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 2 & -4 & -1 \end{pmatrix},$$

are $\lambda_1 = -1$, $\lambda_2 = 1$, and $\lambda_3 = 3$.

[6 marks]

(b) Show that the eigenvectors corresponding to eigenvalues for the matrix A in part (a) are:

$$\underline{v}_1 = (0, 0, 1)^T, \quad \underline{v}_2 = (1, 0, 1)^T, \quad \text{and} \quad \underline{v}_3 = (-2, -2, 1)^T.$$

[9 marks]

(c) Use the Gauss-Seidel method to find the third iteration approximations to x_1, x_2 ,and x_3 for the linear system below. Use the initial approximations $x_1 = 1$, $x_2 = 1$,and $x_3 = 1$.

$$\begin{aligned} 4x_1 + x_2 - x_3 &= 5, \\ 2x_1 + 7x_2 - 3x_3 &= 5, \\ x_1 - 2x_2 + 8x_3 &= -7. \end{aligned}$$

[10 marks]

Formulae

Hyperbolic Formulae

$$\sinh x = \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}, \quad \cosh x = \frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2}.$$

Series Formulae
Taylor's Series

$$\begin{aligned} f(x) &= f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f^{(3)}(a) + \ldots \\ &\quad + \frac{(x-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \ldots \end{aligned}$$

Maclaurin's Series

$$\begin{aligned} f(x) &= f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f^{(3)}(0) + \ldots \\ &\quad + \frac{x^{n-1}}{(n-1)!}f^{(n-1)}(0) + \ldots \end{aligned}$$

or

$$\begin{aligned} y(x) &= y(0) + xy_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \ldots \\ &\quad + \frac{x^{n-1}}{(n-1)!}y_{n-1}(0) + \ldots \end{aligned}$$

Binomial Series

$$(x+y)^n = \left(\begin{array}{c} n \\ 0 \end{array}\right) x^n y^0 + \left(\begin{array}{c} n \\ 1 \end{array}\right) x^{n-1} y^1 + \left(\begin{array}{c} n \\ 2 \end{array}\right) x^{n-2} y^2 + \ldots + \left(\begin{array}{c} n \\ n \end{array}\right) x^0 y^n \qquad \mathrm{n} \in \mathbf{N}$$

Newton-Raphson method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Secant Method

$$x_{n+1} = x_n - f(x_n)\left(\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}\right)$$

Numerical Methods for solution of differential equations

Euler's formula

$$y_{i+1} = y_i + hf(x_i, y_i)$$

Improver Euler method

$$y_{i+1} = y_i + \frac{h}{2}f(x_i, y_i) + \frac{h}{2}f\left(x_{i+1}, y_i + hf(x_i, y_i)\right)$$

Runge-Kutta method of order 4

$$y_{i+1} = y_i + \frac{h}{6}\{k_1 + 2k_2 + 2k_3 + k_4\}$$

$$k_1 = f(x_i, y_i), \qquad k_2 = f(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1),$$

$$k_3 = f(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_2), \qquad k_4 = f(x_i + h, y_i + hk_3).$$

Leibnitz Theorem

$$u_n = \frac{d^n u}{dx^n}, \qquad v_n = \frac{d^n v}{dx^n},$$

$$\frac{d^n(uv)}{dx^n} = \sum_{j=0}^{n} \left(\begin{array}{c} n \\ j \end{array} \right) u_j v_{n-j}$$

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Requirements for this paper:
1. Calculator
2. Graph Paper
3. Log Tables

Question 1.
(a) Show that

$$\sinh(A+B) = \sinh A \cosh B + \cosh A \sinh B,$$

using

$$\cosh x = \frac{\mathrm{e}^{x} + \mathrm{e}^{-x}}{2}, \text{ and } \sinh x = \frac{\mathrm{e}^{x} - \mathrm{e}^{-x}}{2}.$$

[8 marks]

(b) A periodic waveform, $f(t)$, with period 2, that is $f(t+2) = f(t)$, is defined as

$$f(t) = \begin{cases} 1 - t^2 & 0 \leq t < 1 \\ -1 + t & 1 \leq t < 2 \end{cases}.$$

(i) Graph the function on the domain $[0, 6]$.
(ii) Is the function continuous?
(iii) Is the function either even or odd?

[7 marks]

(c) Use Osborne's rule to determine the equivalent hyperbolic form of the following trigonometric identities (briefly justify your answers):

$$\begin{aligned} \text{(i)} \quad 1 + \cos(2t) &= \cos^2 t, \\ \text{(ii)} \quad 2\operatorname{cosec}(2\theta)\cos(2\theta) &= \cot\theta - \tan\theta. \\ \text{(iii)} \quad \sin^3\theta\cos^3\theta &= \frac{3\sin(2\theta) - \sin(6\theta)}{32}, \\ \text{(iv)} \quad \sin(3\theta) &= -4\sin^3(\theta) + 3\sin\theta. \end{aligned}$$

[10 marks]

Question 2.

(a) Show, using the separation of variables method, that the solution to the differential equation

$$\frac{dy}{dx} - (2x\sinh(x^2))y = 2x\sinh(x^2),$$

is

$$y(x) = A\mathrm{e}^{\cosh(x^2)} - 1.$$

Note: A is a constant.

[13 marks]

(b) Use the integrating factor method to show that the solution to the differential equation

$$\frac{dy}{dx} + 2xy = x,$$

is

$$y(x) = \frac{1}{2} + C\mathrm{e}^{-x^2}.$$

[12 marks]

Question 3.

(a) Use the Newton-Raphson method to find the only positive root of the function

$$f(x) = 2\mathrm{e}^{-x} - 1 - x,$$

accurate to 4 places of decimal.

[8 marks]

(b) Show that the Maclaurin series expansion for $f(x) = \mathrm{e}^{-x}$ is

$$\mathrm{e}^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots + \frac{x^{n-1}(-1)^{n-1}}{(n-1)!} + \ldots$$

[6 marks]

(c) Show that the Taylor series expansion for $f(x) = \cos x$ about $x = \frac{\pi}{2}$ (that is in powers of $(x - \frac{\pi}{2})$) is

$$\cos x = -(x - \frac{\pi}{2}) + \frac{(x - \frac{\pi}{2})^3}{3!} - \frac{(x - \frac{\pi}{2})^5}{5!} + \ldots + \frac{(x - \frac{\pi}{2})^{2n-1}}{(2n-1)!}(-1)^n +$$

[7 marks]

(d) Use the first three terms in the expansion from part (c) to approximate $\cos(\frac{5\pi}{8})$.

[4 marks]

Question 4.

(a) Approximate the solution to the initial value problem

$$\frac{dy}{dx} = 2x - y, \quad y(1) = 0,$$

using the Euler Method on the interval $1 < x \leq 2$ where the step size $h = 0.2$. Work throughout to 4 places of decimal.

[14 marks]

(b) Approximate the solution to the initial value problem

$$\frac{dy}{dx} = x^2 \mathrm{e}^{-y}, \quad \text{where } y(1) = 3,$$

at $x = 1.25$ using the Runge-Kutta method of order 4.

[11 marks]

Question 5.

(a) Show, that the indicial equation for the differential equation

$$\frac{d^2y}{dx^2} - \frac{3}{2x}\frac{dy}{dx} + \frac{2}{x}y = 0,$$

is

$$c\left(c - \frac{5}{2}\right)a_0 = 0.$$

[7 marks]

(b) Using the method of Frobenius, show that the solution to the differential equation, given in part (a), is

$$y(x) = A\{1 + \frac{4x}{3} + \frac{8x^2}{3} - \frac{32x^3}{9} + \ldots\} + B\sqrt{x^5}\{1 - \frac{4x}{7} + \frac{8x^2}{63} - \frac{32x^3}{2079} + \ldots\}.$$

[18 marks]

Question 6.

(a) Show that the three distinct eigenvalues of the matrix

$$A = \begin{pmatrix} -2 & 1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix},$$

are $\lambda_1 = -3$, $\lambda_2 = -1$, and $\lambda_3 = 3$.

[7 marks]

(c) Determine the eigenvector corresponding to $\lambda_1 = -3$ for the matrix A given in part (a), show all the mathematical steps.

[5 marks]

(c) Use three iterations of the Jacobi method, with the initial approximation $\underline{x}^{(0)} = (1, 1, 1)$ to find $\underline{x}^{(3)}$ for the linear system:

$$\begin{aligned} 5x_1 + x_2 + 2x_3 &= 19, \\ x_1 + 4x_2 - 2x_3 &= -2, \\ 2x_1 + 3x_2 + 8x_3 &= 39. \end{aligned}$$

Note : the exact solution is $\underline{x} = (2, 1, 4)$.
Why would you expect that the Gauss-Siedel method would converge quicker to the exact solution.

[13 marks]

Formulae

Hyperbolic Formulae

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$$u_n = \frac{d^n u}{dx^n}, \qquad v_n = \frac{d^n v}{dx^n},$$

$$\frac{d^n(uv)}{dx^n} = \sum_{j=0}^{n} \binom{n}{j} u_j v_{n-j}$$